

MySQL
05-Aug-2026

Question

Display the department name along with the highest salary in each department.

Tables

Employees

Emp_ID	Employee_Name	Dept_ID	Salary
101	Asha	1	50000
102	Ravi	2	65000
103	Priya	1	70000
104	Kiran	3	55000
105	Neha	2	80000
106	Arjun	1	62000

Departments

Dept_ID	Department_Name
1	IT
2	HR
3	Finance

Algorithm (How to Think)

Step 1

What information is required?

Department Name

Highest Salary

Step 2

Which table contains the Department Name?

Departments table.

Step 3

Which table contains Salary?

Employees table.

Step 4

Can one table alone solve this?

No.

Need JOIN.

Step 5

How many employees belong to one department?

Many.

So after joining, we must group department-wise.

Step 6

After grouping, how do we get the highest salary?

Use

MAX(Salary)

Step 7

Final Thinking

Join

↓

Group Department

↓

Find MAX Salary

↓

Display Result

SQL Query

```
SELECT
    d.Department_Name,
    MAX(e.Salary) AS Highest_Salary
FROM Employees e
JOIN Departments d
ON e.Dept_ID = d.Dept_ID
GROUP BY d.Department_Name;
```

Output

Department_Name	Highest_Salary
IT	70000
HR	80000
Finance	55000

Step-by-Step Logic

After JOIN

Employee	Department	Salary
Asha	IT	50000
Priya	IT	70000
Arjun	IT	62000
Ravi	HR	65000
Neha	HR	80000
Kiran	Finance	55000

GROUP BY Department

IT

50000
70000
62000

↓

MAX = 70000

HR

65000
80000

↓

MAX = 80000

Finance

55000

↓

MAX = 55000

Final Result

IT	70000
HR	80000
Finance	55000